Earthling Field Guide – Middle School Edition: Teacher Guide

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A browser platformer for grades 6–8 that teaches the 21 PA STEELS Life Science standards (3.1.6-8.A–U) through six worlds of levels, in English and Spanish, plus six just-for-fun pixel bonus speed runs.

About the game

Students play as a friendly alien visiting Earth to study life. It runs in any web browser (no installs, no accounts) and saves progress on each device. There are no written answers: students learn vocabulary and concepts by scanning things, choosing answers at stations, sorting, reading charts and playing mini-games.

- **Language:** English or Spanish (neutral Latin American), chosen with the Language / Idioma buttons on the main menu.
- **Main menu:** a 🎨 Customize your alien button opens the alien choices (body, color, eyes, mouth, antennae, gear) on their own page, with a preview of the 8-bit alien used in the bonus runs. ✓ Done returns to the play choices.
- **Ways to play:** 🎮 Platformer Adventure is the middle school game described in this guide. 🌿 Cozy Exploration, 🛸 Final Mission: Biodome and 🔤 Word Lab came over from the high school game and have not been rebuilt for middle school yet.
- **Other buttons:** Map, 📓 Journal (vocabulary and finds), Progress, Settings (text size, low graphics, music, sound, captions) and About.

The six platformer worlds

There are 26 science levels in six worlds, and every world is open from the start. Each level is a real place or a trip inside a living thing, with stations, sorting games, charts and a level-end Lab Lock.

World	Topic	Levels	Standards (3.1.6-8.)
1	What Living Things Are Made Of	1-1 Pocono Vernal Pool, 1-2 Deep-Sea Vent, 1-3 Backyard Garden, 1-4 Rocky Tide Pool	A, B
2	The Body as a System	2-1 Inside a Hiker's Body, 2-2 Hawk Mountain Trail, 2-3 Cherry Springs at Night, 2-4 Your Brain	C, H
3	Food and Energy for Life	3-1 Jennings Prairie, 3-2 Monterey Kelp Forest, 3-3 A Lunch's Journey (Digestion Mini Golf), 3-4 Garden Test Plots	E, F, G
4	Ecosystems	4-1 Great Barrier Reef, 4-2 Allegheny National Forest, 4-3 Under a Fallen Log, 4-4 Yellowstone After the Fire, 4-5 Susquehanna River Valley (boss: The Silt Storm)	I, J, K, L, U
5	Passing On Traits	5-1 Bowman's Hill Wildflowers, 5-2 Strawberry Coast, 5-3 Cave of the Pale Fish, 5-4 Farm and Research Lab	D, M, N, R
6	Change Over Time	6-1 Canyon Fossil Dig, 6-2 Whale Valley, 6-3 Woods Hole Marine Lab, 6-4 Daphne Major Finches, 6-5 The Selection Simulator (boss: The Graph Gremlin)	O, P, Q, S, T

Students never see standard codes; they appear only on the teacher reference sheet (see Teacher tools).

Bonus speed runs

Each world ends with an optional, just-for-fun 8-bit speed run with no science questions: students collect coins, dodge obstacles and race the clock. They show as square 👾 spaces on the Map and do not count toward any certificate or world achievement. Stars come from the time: beat the par time for 3 stars.

- **The pixel alien:** each student plays as a pixel version of the alien they designed (same shape, color, eyes, mouth, antennae and gear), drawn from the side when moving and from the front when standing. A preview is on the Customize your alien page.
- **Moves:** a glowing orb in each run gives a new move with a section built to practice it. Moves are kept for later bonus runs.
- **Every run also has** moving platforms, wobbly columns that collapse, bricks, patrolling viruses, flying hazards, a crumbling bridge and a boss: a giant virus that must be stomped (or ground-pounded, or spun) 3 times to open the exit gate.

Level	Habitat	New move	How to use it
1-5 Pond Edge	Pond	Ground Pound	In the air, press down: breaks bricks, squashes enemies
2-5 Desert Dash	Desert	Air Dash	In the air, press Shift: one dash per jump
3-5 Rainforest Run	Rainforest	Wall Jump	Jump toward a wall, then jump again while touching it
4-6 Tundra Sprint	Tundra	Glide (plus three sled runs)	While falling, hold jump
5-5 Reef Rush	Coral reef	Spin Attack	Press E: knocks out enemies, small lift
6-6 Penn's Woods	Pennsylvania forest in fall (new for this edition)	All five	A power orb before each section

Penn's Woods shows Appalachian ridges, sugar maples and oaks in fall color, eastern hemlocks (the state tree), mountain laurel (the state flower), old stone walls, a white-tailed deer, a black bear and a wild turkey.

Progress, stars and certificates

Progress is saved in each browser on each device; students who switch computers start over there.

- **Stars (science levels):** 3 stars for 90% or more right on the first try, 2 stars for 70% or more, otherwise 1. A wrong answer adds a few seconds to the time and shows the explanation; after two wrong tries the right choice glows, so no one gets stuck.
- **Stars (bonus runs):** from the time, compared with the par time.
- **Progress screen:** stars, words found and rare specimens for each world, and printable certificates.
- **Certificates:** finishing every science level in a world (at least 1 star each) opens a celebration with confetti and a printable world certificate. Finishing all six worlds earns the whole-game certificate. Bonus runs are not needed. The name typed on a certificate is used only for printing and is never saved.

Teacher tools

- **Teacher link:** add `#teacher` to the end of the game's web address to see the standards reference sheet in About. Add `#level-4-2` (any level number) to jump straight into that level.
- **Admin mode:** on the main menu, type the admin password on the keyboard (nothing appears on screen), or tap the version line at the bottom of the menu 5 times to get a password box. Admin mode unlocks every level, shows the reference sheet, and adds ADMIN to the version line. Do the same again to turn it off. The password is not written in the game's files or on GitHub; only a scrambled code of it is.
- **Teacher: unlock every level:** a switch in Settings that appears in admin mode.
- **Educator reference sheet (About):** every standard, its exact PDE wording, where it is taught and how many items cover it, plus a Print button. Students never see it.
- **Erase progress on this device:** in Settings, for shared classroom computers.

Standards alignment and what is still to come

All 21 grades 6–8 Life Science standards (3.1.6-8.A–U) are taught in the platformer, with standard wording taken word for word from PDE's STEELS document. On November 11, 2026 every item was checked against PDE's STEELS Foundation Boxes (updated September 2025): each standard's Clarifying Statement (what to emphasize) and Assessment Boundary (what the PSSA does not test). All 40 changes from that check are in the game; the full report is PSSA Alignment Check – Middle School Game.

The Grade 8 Science PSSA has four parts. This game covers one of them, Life Science plus Environmental Literacy. The nine Environmental Literacy standards (3.4.6-8.A–I) are planned for the Cozy part of the game and are not taught yet.

Putting a new version on GitHub

The game lives in the Science_Game_PSSA_MS folder on the Desktop. After Claude saves a new version there, open Terminal and paste these two lines, pressing Return after each:

```
cd ~/Desktop/Science_Game_PSSA_MS
git push
```

The version and date show at the bottom of the main menu. Students may need to refresh the page to get a new version.

What changed

Version	Date	Changes
2	Nov 11, 2026	Six bonus speed runs with the pixel alien and five moves, including the new Penn's Woods habitat; 🎨 Customize your alien button on the main menu; 40 changes from the PSSA alignment check (new Punnett square, system-boundary, gene therapy and evidence questions; body systems limited to the six the PSSA tests)
1 (updates)	Oct 25–30, 2026	Worlds 3–6 (17 levels) with Digestion Mini Golf and the Selection Simulator; cavefish question reworded; crumbling-bridge and maze fixes
1	Oct 24, 2026	Worlds 1–2, celebration screen and certificates, Spanish, educator reference sheet

References

Standards and scope documents used to write and check the game:

- Pennsylvania Department of Education. *Science, Technology & Engineering, and Environmental Literacy & Sustainability Standards (STEELS): K-12 Standards.* May 2023. Source of the exact standard wording.
- Pennsylvania Department of Education, Standards Aligned System. *STEELS Foundation Boxes*, grades 6–8 Life Science and Environmental Literacy & Sustainability: Clarifying Statements and Assessment Boundaries. Updated September 2025.
- Pennsylvania Department of Education. *2025 PSSA Science Test Design (Grades 5 & 8, STEELS Aligned)*. 2025.
- Pennsylvania Department of Education. *2025 PSSA Science Grade 8 Item and Scoring Sampler*. 2025. Used for question style only.

The same references appear in the game's About page and on the educator reference sheet.